

IIT JAM 2019 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS

60

TOTAL MARKS

100 M

TEST DURATION

180 Min

PAPER PATTERN

MCQ • MSQ • NAT
(60 Ques)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2019 • Q1 • ID: JAM_2019_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 – Q.10 carry one mark each.Q.1 Let $a_1 = b_1 = 0$, and for each $n \geq 2$, let a_n and b_n be real numbers given by

$$a_n = \sum_{m=2}^n \frac{(-1)^m m}{(\log(m))^m} \text{ and } b_n = \sum_{m=2}^n \frac{1}{(\log(m))^m}.$$

Then which one of the following is TRUE about the sequences $\{a_n\}$ and $\{b_n\}$?

- (A) Both $\{a_n\}$ and $\{b_n\}$ are divergent
- (B) $\{a_n\}$ is convergent and $\{b_n\}$ is divergent
- (C) $\{a_n\}$ is divergent and $\{b_n\}$ is convergent
- (D) Both $\{a_n\}$ and $\{b_n\}$ are convergent

Question 2 (JAM 2019 • Q2 • ID: JAM_2019_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let $T \in M_{m \times n}(\mathbb{R})$. Let V be the subspace of $M_{n \times p}(\mathbb{R})$ defined by

$$V = \{X \in M_{n \times p}(\mathbb{R}) : TX = 0\}.$$

Then the dimension of V is

- (A) $pn - \text{rank}(T)$
- (B) $mn - p \text{rank}(T)$
- (C) $p(m - \text{rank}(T))$
- (D) $p(n - \text{rank}(T))$

Question 3 (JAM 2019 • Q3 • ID: JAM_2019_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. Define $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) = g(x^2 + y^2 - 2z^2).$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

- (A) $4(x^2 + y^2 - 4z^2) g''(x^2 + y^2 - 2z^2)$
- (B) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2)$
- (C) $4(x^2 + y^2 - 2z^2) g''(x^2 + y^2 - 2z^2)$
- (D) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2) + 8g'(x^2 + y^2 - 2z^2)$

Question 4 (JAM 2019 • Q4 • ID: JAM_2019_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be sequences of positive real numbers such that $na_n < b_n < n^2a_n$ for all $n \geq 2$. If the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$ is 4, then the power series $\sum_{n=0}^{\infty} b_n x^n$

- (A) converges for all x with $|x| < 2$
- (B) converges for all x with $|x| > 2$
- (C) does not converge for any x with $|x| > 2$
- (D) does not converge for any x with $|x| < 2$

Question 5 (JAM 2019 • Q5 • ID: JAM_2019_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let S be the set of all limit points of the set $\left\{\frac{n}{\sqrt{2}} + \frac{\sqrt{2}}{n} : n \in \mathbb{N}\right\}$. Let $\mathbb{Q}_+$ be the set of all positive rational numbers. Then

- (A) $\mathbb{Q}_+ \subseteq S$
- (B) $S \subseteq \mathbb{Q}_+$
- (C) $S \cap (\mathbb{R} \setminus \mathbb{Q}_+) \neq \emptyset$
- (D) $S \cap \mathbb{Q}_+ \neq \emptyset$

MA

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Question 6 (JAM 2019 • Q6 • ID: JAM_2019_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 If $x^h y^k$ is an integrating factor of the differential equation

$$y(1 + xy) dx + x(1 - xy) dy = 0,$$

then the ordered pair (h, k) is equal to

- (A) $(-2, -2)$
- (B) $(-2, -1)$
- (C) $(-1, -2)$
- (D) $(-1, -1)$

Question 7 (JAM 2019 • Q7 • ID: JAM_2019_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 If $y(x) = \lambda e^{2x} + e^{\beta x}$, $\beta \neq 2$, is a solution of the differential equation

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

satisfying $\frac{dy}{dx}(0) = 5$, then $y(0)$ is equal to

- (A) 1
- (B) 4
- (C) 5
- (D) 9

Question 8 (JAM 2019 • Q8 • ID: JAM_2019_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 The equation of the tangent plane to the surface $x^2z + \sqrt{8 - x^2 - y^4} = 6$ at the point $(2, 0, 1)$ is

(A) $2x + z = 5$

(B) $3x + 4z = 10$

(C) $3x - z = 10$

(D) $7x - 4z = 10$

Question 9 (JAM 2019 • Q9 • ID: JAM_2019_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 The value of the integral

$$\int_{y=0}^1 \int_{x=0}^{1-y^2} y \sin(\pi(1-x)^2) dx dy$$

is

(A) $\frac{1}{2\pi}$

(B) 2π

(C) $\frac{\pi}{2}$

(D) $\frac{2}{\pi}$

Question 10 (JAM 2019 • Q10 • ID: JAM_2019_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 The area of the surface generated by rotating the curve $x = y^3$, $0 \leq y \leq 1$, about the y -axis, is

(A) $\frac{\pi}{27} 10^{3/2}$

(B) $\frac{4\pi}{3} (10^{3/2} - 1)$

(C) $\frac{\pi}{27} (10^{3/2} - 1)$

(D) $\frac{4\pi}{3} 10^{3/2}$

Q. 11 – Q. 30 carry two marks each.

Question 11 (JAM 2019 • Q11 • ID: JAM_2019_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 Let H and K be subgroups of $\mathbb{Z}_{144}$. If the order of H is 24 and the order of K is 36, then the order of the subgroup $H \cap K$ is

(A) 3

(B) 4

(C) 6

(D) 12

Question 12 (JAM 2019 • Q12 • ID: JAM_2019_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let P be a 4×4 matrix with entries from the set of rational numbers. If $\sqrt{2} + i$, with $i = \sqrt{-1}$, is a root of the characteristic polynomial of P and I is the 4×4 identity matrix, then

- (A) $P^4 = 4P^2 + 9I$ (B) $P^4 = 4P^2 - 9I$ (C) $P^4 = 2P^2 - 9I$ (D) $P^4 = 2P^2 + 9I$

MA

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Question 13 (JAM 2019 • Q13 • ID: JAM_2019_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 The set $\left\{\frac{x}{1+x} : -1 < x < 1\right\}$, as a subset of $\mathbb{R}$, is

- (A) connected and compact
(B) connected but not compact
(C) not connected but compact
(D) neither connected nor compact

Question 14 (JAM 2019 • Q14 • ID: JAM_2019_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 The set $\left\{\frac{1}{m} + \frac{1}{n} : m, n \in \mathbb{N}\right\} \cup \{0\}$, as a subset of $\mathbb{R}$, is

- (A) compact and open (B) compact but not open
(C) not compact but open (D) neither compact nor open

Question 15 (JAM 2019 • Q15 • ID: JAM_2019_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 For $-1 < x < 1$, the sum of the power series $1 + \sum_{n=2}^{\infty} (-1)^{n-1} n^2 x^{n-1}$ is

- (A) $\frac{1-x}{(1+x)^3}$ (B) $\frac{1+x^2}{(1+x)^4}$
(C) $\frac{1-x}{(1+x)^2}$ (D) $\frac{1+x^2}{(1+x)^3}$

Question 16 (JAM 2019 • Q16 • ID: JAM_2019_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 Let $f(x) = (\ln x)^2$, $x > 0$. Then

- (A) $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$ does not exist
(B) $\lim_{x \rightarrow \infty} f'(x) = 2$
(C) $\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = 0$
(D) $\lim_{x \rightarrow \infty} (f(x+1) - f(x))$ does not exist

Question 17 (JAM 2019 • Q17 • ID: JAM_2019_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$. Then $f(1)$ lies in the interval

- (A) $(0, e^{-1})$ (B) $(e^{-1}, \sqrt{e})$ (C) $(\sqrt{e}, e)$ (D) (e, ∞)

Question 18 (JAM 2019 • Q18 • ID: JAM_2019_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 For which one of the following values of k , the equation

$$2x^3 + 3x^2 - 12x - k = 0$$

has three distinct real roots?

- (A) 16 (B) 20 (C) 26 (D) 31

Question 19 (JAM 2019 • Q19 • ID: JAM_2019_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Which one of the following series is divergent?

- (A) $\sum_{n=1}^{\infty} \frac{1}{n} \sin^2 \frac{1}{n}$ (B) $\sum_{n=1}^{\infty} \frac{1}{n} \log n$
(C) $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{1}{n}$ (D) $\sum_{n=1}^{\infty} \frac{1}{n} \tan \frac{1}{n}$

Question 20 (JAM 2019 • Q20 • ID: JAM_2019_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let S be the family of orthogonal trajectories of the family of curves

$$2x^2 + y^2 = k, \text{ for } k \in \mathbb{R} \text{ and } k > 0.$$

If $C \in S$ and C passes through the point $(1, 2)$, then C also passes through

- (A) $(4, -\sqrt{2})$ (B) $(2, -4)$ (C) $(2, 2\sqrt{2})$ (D) $(4, 2\sqrt{2})$

Question 21 (JAM 2019 • Q21 • ID: JAM_2019_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 Let x , $x + e^x$ and $1 + x + e^x$ be solutions of a linear second order ordinary differential equation with constant coefficients. If $y(x)$ is the solution of the same equation satisfying $y(0) = 3$ and $y'(0) = 4$, then $y(1)$ is equal to

- (A) $e + 1$ (B) $2e + 3$ (C) $3e + 2$ (D) $3e + 1$

Question 22 (JAM 2019 • Q22 • ID: JAM_2019_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 The function

$$f(x, y) = x^3 + 2xy + y^3$$

has a saddle point at

- (A) $(0, 0)$ (B) $\left(-\frac{2}{3}, -\frac{2}{3}\right)$ (C) $\left(-\frac{3}{2}, -\frac{3}{2}\right)$ (D) $(-1, -1)$

Question 23 (JAM 2019 • Q23 • ID: JAM_2019_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 The area of the part of the surface of the paraboloid $x^2 + y^2 + z = 8$ lying inside the cylinder $x^2 + y^2 = 4$ is

- (A) $\frac{\pi}{2}(17^{3/2} - 1)$ (B) $\pi(17^{3/2} - 1)$ (C) $\frac{\pi}{6}(17^{3/2} - 1)$ (D) $\frac{\pi}{3}(17^{3/2} - 1)$

Question 24 (JAM 2019 • Q24 • ID: JAM_2019_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 Let $\mathcal{C}$ be the circle $(x - 1)^2 + y^2 = 1$, oriented counter clockwise. Then the value of the line integral

$$\oint_{\mathcal{C}} -\frac{4}{3}xy^3 dx + x^4 dy$$

is

(A) 6π (B) 8π (C) 12π (D) 14π

Question 25 (JAM 2019 • Q25 • ID: JAM_2019_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $\vec{F}(x, y, z) = 2y\hat{i} + x^2\hat{j} + xy\hat{k}$ and let $\mathcal{C}$ be the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 1$. Then the value of

$$\left| \oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} \right|$$

is

(A) π (B) $\frac{3\pi}{2}$ (C) 2π (D) 3π

MA

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Question 26 (JAM 2019 • Q26 • ID: JAM_2019_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The tangent line to the curve of intersection of the surface $x^2 + y^2 - z = 0$ and the plane $x + z = 3$ at the point $(1, 1, 2)$ passes through

(A) $(-1, -2, 4)$ (B) $(-1, 4, 4)$ (C) $(3, 4, 4)$ (D) $(-1, 4, 0)$

Question 27 (JAM 2019 • Q27 • ID: JAM_2019_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The set of eigenvalues of which one of the following matrices is NOT equal to the set of eigenvalues of $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$?

(A) $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

(B) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

Question 28 (JAM 2019 • Q28 • ID: JAM_2019_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Let $\{a_n\}$ be a sequence of positive real numbers. The series $\sum_{n=1}^{\infty} a_n$ converges if the series

(A) $\sum_{n=1}^{\infty} a_n^2$ converges

(B) $\sum_{n=1}^{\infty} \frac{a_n}{2^n}$ converges

(C) $\sum_{n=1}^{\infty} \frac{a_{n+1}}{a_n}$ converges

(D) $\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}}$ converges

Question 29 (JAM 2019 • Q29 • ID: JAM_2019_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 For $\beta \in \mathbb{R}$, define

$$f(x, y) = \begin{cases} \frac{x^2 |x|^\beta y}{x^4 + y^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then, at $(0, 0)$, the function f is

(A) continuous for $\beta = 0$ (B) continuous for $\beta > 0$ (C) not differentiable for any β (D) continuous for $\beta < 0$

Question 30 (JAM 2019 • Q30 • ID: JAM_2019_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $\{a_n\}$ be a sequence of positive real numbers such that

$$a_1 = 1, a_{n+1}^2 - 2a_n a_{n+1} - a_n = 0 \text{ for all } n \geq 1.$$

Then the sum of the series $\sum_{n=1}^{\infty} \frac{a_n}{3^n}$ lies in the interval

- (A) $(1, 2]$ (B) $(2, 3]$ (C) $(3, 4]$ (D) $(4, 5]$

MA

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Question 31 (JAM 2019 • Q31 • ID: JAM_2019_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let G be a noncyclic group of order 4. Consider the statements I and II:

- I. There is NO injective (one-one) homomorphism from G to $\mathbb{Z}_8$
II. There is NO surjective (onto) homomorphism from $\mathbb{Z}_8$ to G

Then

- (A) I is true (B) I is false
(C) II is true (D) II is false

Question 32 (JAM 2019 • Q32 • ID: JAM_2019_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 Let G be a nonabelian group, $y \in G$, and let the maps f, g, h from G to itself be defined by

$$f(x) = yxy^{-1}, \quad g(x) = x^{-1} \quad \text{and} \quad h = g \circ g.$$

Then

- (A) g and h are homomorphisms and f is not a homomorphism
- (B) h is a homomorphism and g is not a homomorphism
- (C) f is a homomorphism and g is not a homomorphism
- (D) f, g and h are homomorphisms

Question 33 (JAM 2019 • Q33 • ID: JAM_2019_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let S and T be linear transformations from a finite dimensional vector space V to itself such that $S(T(v)) = 0$ for all $v \in V$. Then

- (A) $\text{rank}(T) \geq \text{nullity}(S)$
- (B) $\text{rank}(S) \geq \text{nullity}(T)$
- (C) $\text{rank}(T) \leq \text{nullity}(S)$
- (D) $\text{rank}(S) \leq \text{nullity}(T)$

Question 34 (JAM 2019 • Q34 • ID: JAM_2019_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 Let $\vec{F}$ and $\vec{G}$ be differentiable vector fields and let g be a differentiable scalar function. Then

- (A) $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot \nabla \times \vec{F} - \vec{F} \cdot \nabla \times \vec{G}$
- (B) $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot \nabla \times \vec{F} + \vec{F} \cdot \nabla \times \vec{G}$
- (C) $\nabla \cdot (g\vec{F}) = g\nabla \cdot \vec{F} - \nabla g \cdot \vec{F}$
- (D) $\nabla \cdot (g\vec{F}) = g\nabla \cdot \vec{F} + \nabla g \cdot \vec{F}$

Question 35 (JAM 2019 • Q35 • ID: JAM_2019_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Consider the intervals $S = (0, 2]$ and $T = [1, 3)$. Let S° and T° be the sets of interior points of S and T , respectively. Then the set of interior points of $S \setminus T$ is equal to

- (A) $S \setminus T^\circ$
- (B) $S \setminus T$
- (C) $S^\circ \setminus T^\circ$
- (D) $S^\circ \setminus T$

Question 36 (JAM 2019 • Q36 • ID: JAM_2019_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Let $\{a_n\}$ be the sequence given by

$$a_n = \max\left\{\sin\left(\frac{n\pi}{3}\right), \cos\left(\frac{n\pi}{3}\right)\right\}, \quad n \geq 1.$$

Then which of the following statements is/are TRUE about the subsequences $\{a_{6n-1}\}$ and $\{a_{6n+4}\}$?

- (A) Both the subsequences are convergent
- (B) Only one of the subsequences is convergent
- (C) $\{a_{6n-1}\}$ converges to $-\frac{1}{2}$
- (D) $\{a_{6n+4}\}$ converges to $\frac{1}{2}$

MA

8/13

Question 37 (JAM 2019 • Q37 • ID: JAM_2019_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let

$$f(x) = \cos(|\pi - x|) + (x - \pi) \sin |x| \text{ and } g(x) = x^2 \text{ for } x \in \mathbb{R}.$$

If $h(x) = f(g(x))$, then

- (A) h is not differentiable at $x = 0$
- (B) $h'(\sqrt{\pi}) = 0$
- (C) $h''(x) = 0$ has a solution in $(-\pi, \pi)$
- (D) there exists $x_0 \in (-\pi, \pi)$ such that $h(x_0) = x_0$

Question 38 (JAM 2019 • Q38 • ID: JAM_2019_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let $f: \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by

$$f(x) = (\sin x)^\pi - \pi \sin x + \pi.$$

Then which of the following statements is/are TRUE?

- (A) f is an increasing function
- (B) f is a decreasing function
- (C) $f(x) > 0$ for all $x \in \left(0, \frac{\pi}{2}\right)$
- (D) $f(x) < 0$ for some $x \in \left(0, \frac{\pi}{2}\right)$

Question 39 (JAM 2019 • Q39 • ID: JAM_2019_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let

$$f(x, y) = \begin{cases} \frac{|x|}{|x| + |y|} \sqrt{x^4 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

Then at $(0, 0)$,

- (A) f is continuous
 (B) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y}$ does not exist
 (C) $\frac{\partial f}{\partial x}$ does not exist and $\frac{\partial f}{\partial y} = 0$
 (D) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

Question 40 (JAM 2019 • Q40 • ID: JAM_2019_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $\{a_n\}$ be the sequence of real numbers such that

$$a_1 = 1 \text{ and } a_{n+1} = a_n + a_n^2 \text{ for all } n \geq 1.$$

Then

- (A) $a_4 = a_1(1 + a_1)(1 + a_2)(1 + a_3)$ (B) $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$
 (C) $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 1$ (D) $\lim_{n \rightarrow \infty} a_n = 0$

MA

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Question 41 (JAM 2019 • Q41 • ID: JAM_2019_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 Let x be the 100-cycle $(1 \ 2 \ 3 \ \cdots \ 100)$ and let y be the transposition $(49 \ 50)$ in the permutation group S_{100} . Then the order of xy is _____

Question 42 (JAM 2019 • Q42 • ID: JAM_2019_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let W_1 and W_2 be subspaces of the real vector space $\mathbb{R}^{100}$ defined by

$$W_1 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 4 \},$$

$$W_2 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 5 \}.$$

Then the dimension of $W_1 \cap W_2$ is _____

Question 43 (JAM 2019 • Q43 • ID: JAM_2019_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the following system of three linear equations in four unknowns x_1, x_2, x_3 and x_4

$$x_1 + x_2 + x_3 + x_4 = 4,$$

$$x_1 + 2x_2 + 3x_3 + 4x_4 = 5,$$

$$x_1 + 3x_2 + 5x_3 + kx_4 = 5.$$

If the system has no solutions, then $k =$ _____

Question 44 (JAM 2019 • Q44 • ID: JAM_2019_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let $\vec{F}(x, y) = -y \hat{i} + x \hat{j}$ and let C be the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

oriented counter clockwise. Then the value of $\oint_C \vec{F} \cdot d\vec{r}$ (round off to 2 decimal places) is _____

Question 45 (JAM 2019 • Q45 • ID: JAM_2019_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 The coefficient of $\left(x - \frac{\pi}{2}\right)$ in the Taylor series expansion of the function

$$f(x) = \begin{cases} \frac{4(1 - \sin x)}{2x - \pi}, & x \neq \frac{\pi}{2} \\ 0, & x = \frac{\pi}{2} \end{cases}$$

about $x = \frac{\pi}{2}$, is _____

MA

10/13

Question 46 (JAM 2019 • Q46 • ID: JAM_2019_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $f: [0, 1] \rightarrow \mathbb{R}$ be given by

$$f(x) = \frac{\left(1+x^{\frac{1}{3}}\right)^3 + \left(1-x^{\frac{1}{3}}\right)^3}{8(1+x)}.$$

Then

$$\max \{f(x): x \in [0, 1]\} - \min \{f(x): x \in [0, 1]\}$$

is _____

Question 47 (JAM 2019 • Q47 • ID: JAM_2019_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 If

$$g(x) = \int_{x(x-2)}^{4x-5} f(t) dt, \text{ where } f(x) = \sqrt{1+3x^4} \text{ for } x \in \mathbb{R}$$

then $g'(1) =$ _____

Question 48 (JAM 2019 • Q48 • ID: JAM_2019_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let

$$f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 - y^2}, & x^2 - y^2 \neq 0 \\ 0, & x^2 - y^2 = 0. \end{cases}$$

Then the directional derivative of f at $(0, 0)$ in the direction of $\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$ is _____

Question 49 (JAM 2019 • Q49 • ID: JAM_2019_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 The value of the integral

$$\int_{-1}^1 \int_{-1}^1 |x + y| dx dy$$

(round off to 2 decimal places) is _____

Question 50 (JAM 2019 • Q50 • ID: JAM_2019_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 The volume of the solid bounded by the surfaces $x = 1 - y^2$ and $x = y^2 - 1$, and the planes $z = 0$ and $z = 2$ (round off to 2 decimal places) is _____

Q. 51 – Q. 60 carry two marks each

Question 51 (JAM 2019 • Q51 • ID: JAM_2019_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 The volume of the solid of revolution of the loop of the curve $y^2 = x^4(x + 2)$ about the x -axis (round off to 2 decimal places) is _____

Question 52 (JAM 2019 • Q52 • ID: JAM_2019_Q52)**NAT****+2 M****Neg: Nil (0)**

Q.52 The greatest lower bound of the set

$$\{(e^n + 2^n)^{\frac{1}{n}} : n \in \mathbb{N}\},$$

(round off to 2 decimal places) is _____

Question 53 (JAM 2019 • Q53 • ID: JAM_2019_Q53)**NAT****+2 M****Neg: Nil (0)**

Q.53 Let $G = \{n \in \mathbb{N} : n \leq 55, \gcd(n, 55) = 1\}$ be the group under multiplication modulo 55.
Let $x \in G$ be such that $x^2 = 26$ and $x > 30$. Then x is equal to _____

Question 54 (JAM 2019 • Q54 • ID: JAM_2019_Q54)**NAT****+2 M****Neg: Nil (0)**

Q.54 The number of critical points of the function

$$f(x, y) = (x^2 + 3y^2)e^{-(x^2+y^2)}$$

is _____

Question 55 (JAM 2019 • Q55 • ID: JAM_2019_Q55)**NAT****+2 M****Neg: Nil (0)**

Q.55 The number of elements in the set $\{x \in S_3 : x^4 = e\}$, where e is the identity element of the permutation group S_3 , is _____

Question 56 (JAM 2019 • Q56 • ID: JAM_2019_Q56)**NAT****+2 M****Neg: Nil (0)**

Q.56 If $\begin{pmatrix} 2 \\ y \\ z \end{pmatrix}$, $y, z \in \mathbb{R}$, is an eigenvector corresponding to a real eigenvalue of the matrix $\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -4 \\ 0 & 1 & 3 \end{pmatrix}$
then $z - y$ is equal to _____

Question 57 (JAM 2019 • Q57 • ID: JAM_2019_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let M and N be any two 4×4 matrices with integer entries satisfying

$$MN = 2 \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the maximum value of $\det(M) + \det(N)$ is _____

Question 58 (JAM 2019 • Q58 • ID: JAM_2019_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let M be a 3×3 matrix with real entries such that $M^2 = M + 2I$, where I denotes the 3×3 identity matrix. If α, β and γ are eigenvalues of M such that $\alpha\beta\gamma = -4$, then $\alpha + \beta + \gamma$ is equal to _____

Question 59 (JAM 2019 • Q59 • ID: JAM_2019_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let $y(x) = xv(x)$ be a solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 3y = 0.$$

If $v(0) = 0$ and $v(1) = 1$, then $v(-2)$ is equal to _____

MA

12/13

Question 60 (JAM 2019 • Q60 • ID: JAM_2019_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x)$ is the solution of the initial value problem

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0, \quad y(0) = 2, \quad \frac{dy}{dx}(0) = 0,$$

then $y(\ln 2)$ is (round off to 2 decimal places) equal to _____

END OF THE QUESTION PAPER

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2019 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	D
2	MCQ	+1	D
3	MCQ	+1	B
4	MCQ	+1	A
5	MCQ	+1	B
6	MCQ	+1	A
7	MCQ	+1	C
8	MCQ	+1	B
9	MCQ	+1	A
10	MCQ	+1	C
11	MCQ	+2	D
12	MCQ	+2	C
13	MCQ	+2	B
14	MCQ	+2	B
15	MCQ	+2	A
16	MCQ	+2	C
17	MCQ	+2	D
18	MCQ	+2	A
19	MCQ	+2	B
20	MCQ	+2	C

Q. No.	Type	Marks	Official Key
21	MCQ	+2	D
22	MCQ	+2	A
23	MCQ	+2	C
24	MCQ	+2	B
25	MCQ	+2	C
26	MCQ	+2	B
27	MCQ	+2	D
28	MCQ	+2	C
29	MCQ	+2	B
30	MCQ	+2	A
31	MSQ	+2	A, C
32	MSQ	+2	B, C
33	MSQ	+2	C, D
34	MSQ	+2	A, D
35	MSQ	+2	B, D
36	MSQ	+2	A
37	MSQ	+2	B, C, D
38	MSQ	+2	B, C
39	MSQ	+2	A, D
40	MSQ	+2	A, B

Q. No.	Type	Marks	Official Key
41	NAT	+1	99 to 99
42	NAT	+1	60 to 60
43	NAT	+1	7 to 7
44	NAT	+1	75.35 to 75.45
45	NAT	+1	1 to 1
46	NAT	+1	0.25 to 0.25
47	NAT	+1	8 to 8
48	NAT	+1	2.6 to 2.6
49	NAT	+1	2.60 to 2.70
50	NAT	+1	5.30 to 5.50
51	NAT	+2	6.60 to 6.80
52	NAT	+2	2.69 to 2.74
53	NAT	+2	31 to 31 or 46 to 46
54	NAT	+2	5 to 5
55	NAT	+2	4 to 4
56	NAT	+2	3 to 3
57	NAT	+2	17 to 17
58	NAT	+2	3 to 3
59	NAT	+2	4 to 4
60	NAT	+2	1.12 to 1.25

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.